COMSATS University Islamabad, Attock Campus

Department of Computer Science

Course:

DATA STRUCTUES

Lab Project

Café Management System

Submitted To

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Project Report:

Café Management System

1. Introduction

A Cafe Management System is a software application used to organize and manage cafe activities efficiently. In a busy cafe environment, different types of customers and services must be handled properly. Some customers prefer dine-in service, some choose take-away, and others request home delivery.

- Different operations require different processing techniques. For example:
- Dine-in customers should be served in First-In First-Out order.
- Home delivery orders may use Last-In First-Out processing.
- Take-away customers may require priority-based service.
- Customer records should support fast searching and sorting.

This project demonstrates how theoretical concepts of data structures can be applied in a real-world application.

2. Project Objective

The objective of this project is to design and implement a Cafe Management System in JAVA using multiple data structures in Java. The system simulates real-world cafe operations including take-away orders, dine-in orders, home delivery services, served customer records, and customer feedback management.

This project demonstrates the practical implementation of different data structures including Linked Lists, Stacks, Queues, Heap, AVL Tree, and Binary Search Tree (BST). Each data structure is selected according to the functionality where it performs most efficiently.

The system is fully menu-driven and allows users to:

- Place different types of orders
- Serve customers according to rules
- Search and manage records efficiently
- Display pending and served orders
- Generate reports and earnings
- Handle customer feedback

The project also helps students understand how data structures are applied in real-life management systems.

3. Data Structures Used:

i. Singly Linked List

<u>Purpose:</u> Used for Dine-In customer queue. Because Linked List allows dynamic memory allocation and efficient insertion of customers.

<u>Working:</u>

New customers are inserted at the end.

Customers are served from the front.

ii. Queue (FIFO)

<u>Purpose:</u> Used for Dine-In order processing. Because Dine-in customers should be served in the same order in which they arrive.

<u>Example</u>: If customers arrive in this order:

Ali, Ahmed, Sara

Then they are served in the same order.

iii. Stacks (LIFO)

Purpose: Used for Home Delivery order processing. Because most recent delivery orders can be dispatched first.
Example: If delivery orders are:
Ali
Ahmed
Sara
Then Sara will be dispatched first.

iv. MAX Heap

Purpose: Used for Take-Away customer priority handling. Because Heap efficiently handles priority-based processing.
Older customer age = Higher priority
Example: If customers are
Ali : 45
Ahmed : 39
Sara : 26
Ahmed will be served first.

v. AVL TREE

Purpose: Used for storing served customer records. Because AVL Tree is a self-balancing Binary Search Tree that provides fast searching, insertion, and deletion.
Key Used:
Customer Name + Unique ID

vi. Binary Search Tree (BST)

Purpose: Used for customer feedback management. Because Feedback is sorted according to customer ratings.
Working:
Lower ratings go to left subtree.
Higher ratings go to right subtree.
Benefit
In-order traversal displays feedback from lowest to highest rating.

4. Core Functionalities Implemented

a) Add New Record

The system allows adding:
- Take-away orders
- Dine-in orders
- Home delivery orders
- Customer feedback

Data Structures Used:
Heap, Queue, Stack, BST

b) Search for a Record

The system searches served customers using AVL Tree.
Searching Method:
Customer name is searched recursively in AVL Tree.
Data Structure Used:
AVL Tree

c) Update/Delete Record

The system supports deletion of served customer records.

Data Structure Used:

AVL Tree

d) Display All Records

The system displays:

- Pending take-away orders
- Pending dine-in orders
- Pending delivery orders
- Served customers
- Customer feedback

Data Structures Used:

Heap, Queue, Stack, AVL Tree, BST

e) Generate Report and Statistics

The system generates:

- Total pending bill
- Total earnings
- Customer feedback reports

Data Structures Used:

AVL Tree and BST

f) Priority Based Processing

The system processes take-away customers using Max Heap.

- Priority Rule

Older customers are served first.

Data Structure Used:

Max Heap

g) History Tracking / Stack Processing

Home delivery orders are processed using Stack.

Working:

Most recently added order is dispatched first.

Data Structure Used:

Stack using Linked List

5. Working of the System

- Take-Away Orders

Orders are inserted into Max Heap.
Customer age determines priority.
Oldest customer is served first.

- Dine-In Orders

Orders are inserted into Queue.
Customers are served in arrival order.

- Home Delivery Orders

Orders are pushed into Stack.
Latest order is dispatched first.

- Served Customers

All served customers are stored in AVL Tree.
Records are stored alphabetically.
Searching becomes efficient.

- Customer Feedback

Feedback is inserted into BST.
Reviews are sorted according to ratings.

6. Sample Inputs/Output

IMPLEMENTATION

CAFÉ MANAGEMENT SYSTEM OUTPUT DASHBOARD

When the program starts, the system displays a welcome dashboard of **Arooj's Cafe** along with the cafe address and complete coffee menu with prices. After that, an interactive main menu appears where the user can place orders, serve customers, manage records, view feedback, and perform different cafe management operations using various data structures.

```
PS D:\BSE-4 RAFIA\Data Structures\DS-PROJECT>  & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExce
onMessages' '-cp' 'C:\Users\Discount Laptop\AppData\Roaming\Code\User\workspaceStorage\55ee0853c336f57bcb32454ee79b1361\
hat.java\jdt_ws\DS-PROJECT_f41712\bin' 'CafeManagementSystem'
==============================================
      Welcome to Arooj's Cafe
      COMSATS University Islamabad, Attock Campus
==============================================
MENU:
   1. Espresso              Rs. 300
   2. Double Espresso       Rs. 450
   3. Latte                 Rs. 400
   4. Americano             Rs. 350
   5. Macchiato             Rs. 500
   6. Flat White            Rs. 480
   7. Cappuccino            Rs. 420
   8. Matcha                Rs. 380
   9. Masala Chai           Rs. 200
  10. V60 Brew              Rs. 550
==============================================

========= MAIN MENU =========
--- PLACE ORDER ---
 1. Take-Away Order
 2. Dine-In Order
 3. Home Delivery Order
--- SERVE ORDER ---
 4. Serve Take-Away (Max-Heap)
 5. Serve Dine-In   (Queue)
 6. Serve Delivery  (Stack)
 7. Serve ALL Orders
--- DISPLAY PENDING ---
 8. Show Take-Away Orders
 9. Show Dine-In Orders
10. Show Delivery Orders
11. Total Bill of Pending Orders
--- SERVED RECORDS (AVL Tree) ---
12. Show All Served Customers
13. Search Served Customer by Name
14. Delete a Served Record
15. Clear All Served Records
16. Total Earnings
--- FEEDBACK (BST) ---
17. Submit Feedback
18. View All Feedback
--- EXIT ---
 0. Exit
=============================
Enter your choice:
```

Take Away Order (MAX-HEAP)

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

============================
Enter your choice: 1

Customer Name  : RAFIYA
Customer Age   : 18
Quantity       : 2
Coffee (1-10)  : 6
Take-Away order placed for: RAFIYA (Age: 18) 1 Rs. 960.0
```

Dine-In-Order (QUEUE)

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Enter your choice: 2

Customer Name  : SAMRINA
Customer Age   : 19
Quantity       : 1
Coffee (1-10)  : 3
Dine-In order placed for: SAMRINA  Rs. 400.0
```

Home Delivery Order (STACK)

```
============================
Enter your choice: 3

Customer Name  : Sara
Customer Age   : 30
Quantity       : 2
Coffee (1-10)  : 5
Delivery Address: Kamra City
Distance (km)  : 4
Home Delivery order placed for: Sara to Kamra  Rs. 1200.0 (incl. delivery charges Rs. 200.0)
```

Serve Take Away Customer

```
Enter your choice: 4

Served Take-Away: rafia (Age: 18)
```

Serve Home Delivery Customer

```
Enter your choice: 6

Dispatched Home Delivery: Sara to Kamra
```

Display Take-Away Orders

```
Enter your choice: 8

No take-away orders pending.
```

Display Dine-In Orders

```
Enter your choice: 9

—— Dine-In Pending Orders ——
  Name     : samrina
  Age      : 29
  Coffee   : Macchiato
  Quantity : 3
  Bill     : Rs. 1500.0
```

Display Home Delivery Orders

```
Enter your choice: 10

No home delivery orders pending.
```

Total Pending Bill

```
Enter your choice: 11
________________________________

  Take-Away pending total   : Rs. 0.0
  Dine-In pending total     : Rs. 1500.0
  Home Delivery pending total: Rs. 0.0
  Grand total               : Rs. 1500.0
```

Show All Served Customers (AVL Tree)

```
Enter your choice: 12
________________________________

—— Served Customers (alphabetical) ——
  Name         : Sara
  Age          : 30
  Coffee       : Macchiato
  Quantity     : 2
  Bill         : Rs. 1200.0
  Order Type   : Home-Delivery
  ________________________________

  Name         : rafia
  Age          : 18
  Coffee       : Flat White
  Quantity     : 2
  Bill         : Rs. 960.0
  Order Type   : Take-Away
  ________________________________
```

Search Served Customer

```
Enter your choice: 13
________________________________

Enter customer name to search: SARA
Customer not found in served records.
```

Delete Served Customer Record

```
Enter your choice: 14
________________________________

Enter customer name to delete: RAFIA
Record deleted (if it existed).
```

Clear All Served Records

```
Enter your choice: 15
________________________________

Served customers list cleared.
```

Total Earnings

```
Enter your choice: 16
________________________________

  Total Earnings: Rs. 2900.00
```

View All Feedback

```
Enter your choice: 17
________________________________

Your Name   : RAFIA
Rating (1-5): 5
Your Review : AMAZING
Thank you for your feedback!
```

Exit Program

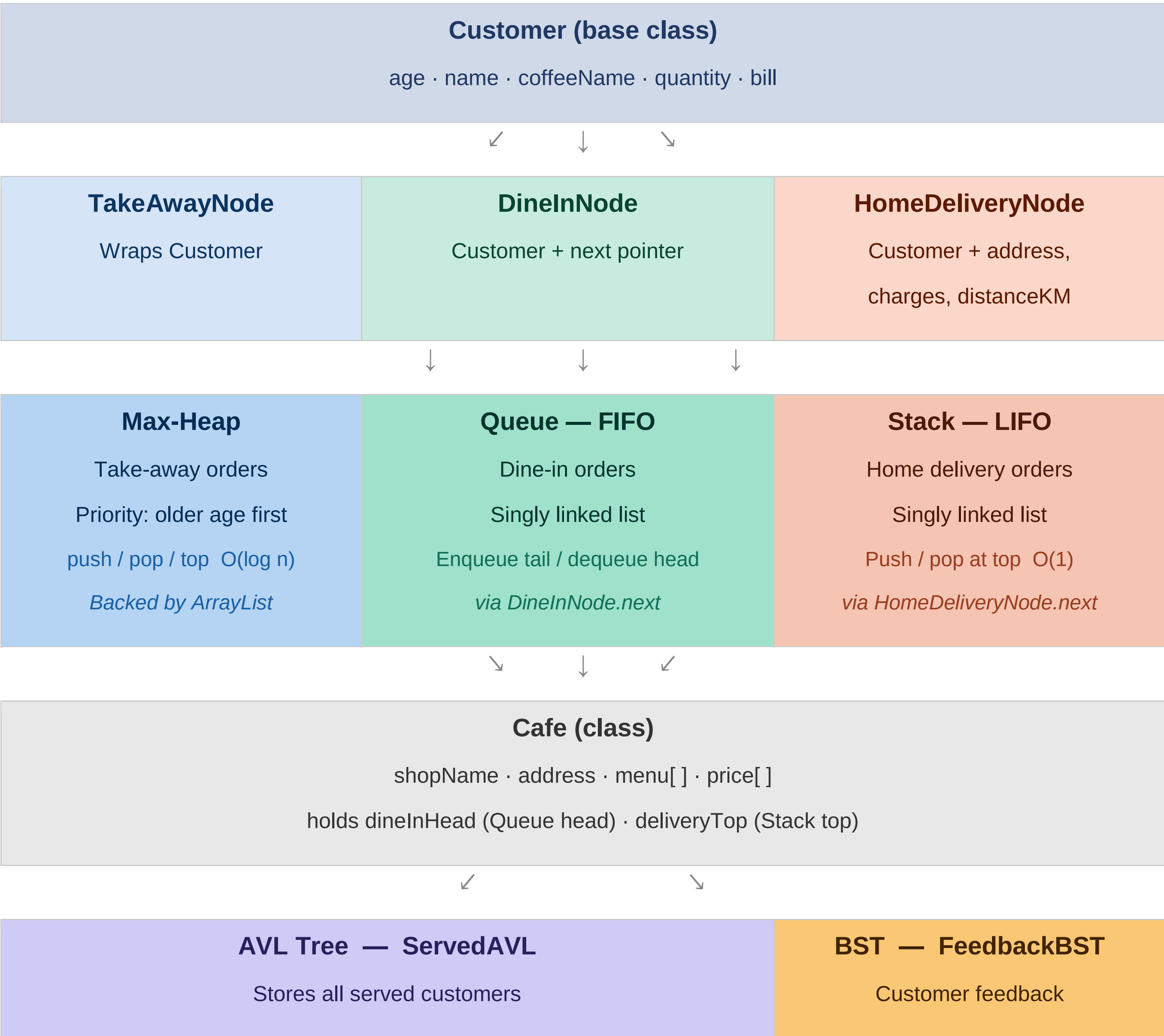
```
==============================
Enter your choice: 0
________________________________

Thank you for visiting Arooj's Cafe! Goodbye!
PS D:\BSE-4 RAFIA\Data Structures\DS-PROJECT>
```

7. Problem Faced During Development

Duplicate Customer Names in AVL Tree

Problem:
AVL Tree does not allow duplicate keys. If the same customer name was inserted again, the system generated an error.
Example:
Rafia already exists
Solution:
A unique ID counter was added with customer names.
Example:
Rafia_1
Rafia_2
Rafia_3
This solved duplicate insertion issues.

8. Flow Chart



Key: customer name (alphabetical) LL / RR / LR / RL rotations — O(log n) *insert · search · delete · inorder · totalEarnings*	Key: rating (1–5 stars) Lower → left | Higher → right *insert · inorder display O(log n)*

↘ ↙

CafeManagementSystem (main)

Menu-driven user interface via Scanner (java.util.Scanner)

Manages all DS: MaxHeap · Cafe (Queue/Stack) · ServedAVL · FeedbackBST

Take-away (Max-Heap)	**Dine-in (Queue / SLL)**	**Home delivery (Stack / SLL)**	**Served records (AVL tree)**
Feedback (BST)		**Shared / Utility (Cafe, Customer, Main)**	